Chapter 19 Notes

Section 2

Notes

For the model in this section to work, we need $\omega_2 \in M$. However, it is not clear to my why this is.

If we assume that $\omega_2 \in M$, here is why $\langle \mathcal{G}, \mathcal{R} \rangle$ is an M-set. It is easy to see that each particular forcing condition is in M - each particular one can be explicitly defined, as it is assumed that $p(\alpha) = \langle \emptyset, \emptyset \rangle$ for all but a finite number of $\alpha \in \omega_2$. The set $\mathcal{G}$ of all forcing conditions is a subset of $\mathcal{P}(\mathcal{P}(\omega_2 \times (\mathcal{P}(\omega \times \omega) \cap M)) \cap M) \cap M$, which is in M. $\mathcal{G}$ is also easily seen to be definable over M, and consequently, $\mathcal{G} \in M$. Then $\mathcal{R}$ is also easily seen to be in M, as it is definable over M, and is a subset of $\mathcal{G} \times \mathcal{G}$, which is in M (as we have shown $\mathcal{G}$ to be in M).

Exercise 2.1

This is basically the same proof as that of Theorem 4.1 of Chapter 18.

Proof. We prove that, for each $\alpha \in \omega_2$, $[\![\sim\text{Const}(f_\alpha)]\!]$ is valid in the model M. Suppose, to the contrary, that $p \nvDash [\![\sim\text{Const}(f_\alpha)]\!]$ for some p. Then for some p' with $p\mathcal{R}p'$, $p' \vDash [\![\text{Const}(f_\alpha)]\!]$. By Theorem 3.2 of Chapter 18, it follows that $p \vDash \Diamond [\![f_\alpha \approx \hat{c}]\!]$ for some contrsuctible set $c \in M$. This means that, for some r with $p'\mathcal{R}r$, $r \vDash [\![f_\alpha \approx \hat{c}]\!]$. We derive a contradiction from this. r is some forcing condition, so suppose that $r(\alpha) = \langle P, N \rangle$. There are then two cases.

Case 1) $c \subseteq P$. Because both P and N are finite, we can pick some integer n such that $n \notin P \cup N$ (thus also $n \notin c$). Let $P' = P \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P', N \rangle$. Obviously $r\mathcal{R}q$. Because $c \in P'$, it follows that $q \vDash [\![\hat{n} \in f_\alpha]\!]$. But because $n \notin c$, $q \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$. These two things jointly imply $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$, as $r \vDash [\![f_\alpha \approx \hat{c}]\!]$ and $r\mathcal{R}q$. Thus we have our contradiction in this case.

Case 2) $c \nsubseteq P$. Pick some n in c but not in P. Let $N' = N \cup \{n\}$. Then there is some forcing condition q such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = r(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P, N' \rangle$. Obviously $r\mathcal{R}q$. Because $n \in c$, it follows that $q \vDash [\![\hat{n} \in \hat{c}]\!]$. But because $n \in N'$, it follows that $q \vDash [\![\sim(\hat{n} \in f_\alpha)]\!]$ (Reason: Suppose, for some q' with $q\mathcal{R}q'$, that $q' \vDash \hat{n} \in f_\alpha$. Then for some x, $q' \vDash [\![x \epsilon f_\alpha]\!]$ and $q' \vDash [\![x \approx \hat{n}]\!]$. This means x is $\hat{a}$ for some a, and thus, by Lemma 3.2 of Chapter 17, a and n are the same set. But this means $q' \vDash [\![\hat{n} \epsilon f_\alpha]\!]$, which cannot happen, as $n \in N'$). These two things imply that $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$ (for the same reason as in the first case), which is our contradiction.

□

Section 3

Exercise 3.1

We prove that $[\![f : \hat{\omega} \xrightarrow[\text{onto}]{\text{1-1}} \hat{\omega_2}]\!]$ is valid in the described $S4$-regular model described. Consider any artibrary world p. It suffices to show that the following four things hold:

$$(1)\ p \vDash [\![(\forall x)(\forall y)(\forall z)((\langle x, y\rangle \in f \wedge \langle x, z\rangle \in f) \supset y = z)]\!]$$

(this is the translate of the formula saying f is a function)

$$(2)\ p \vDash [\![(\forall x)(\forall y)(\forall z)((\langle x, z\rangle \in f \wedge \langle y, z\rangle \in f) \supset x = y)]\!]$$

(this is the translate of the formula saying that f is 1-1)

$$(3)\ p \vDash [\![(\forall x)(x \in \hat{\omega} \supset (\exists y)(y \in \widehat{\omega_2} \wedge \langle x, y\rangle \in f))]\!]$$

(this is the translate of the formula which says that, for each $y \in \omega$, there is some $x \in \omega_2$ such that $f(x) = y$)

$$(4)\ p \vDash [\![(\forall x)(x \in \widehat{\omega_2} \supset (\exists y)(y \in \hat{\omega} \wedge \langle x, y\rangle \in f))]\!]$$

(this is the translate of the formula which says that, for each $x \in \omega_2$, there is some $y \in \omega$ such that $f(x) = y$)

(1) We first show that formula (1) is true at p. It suffices to consider any a, b, c, and show that $p \vDash [\![((\langle a, b\rangle \in f \wedge \langle a, c\rangle \in f) \supset b = c)]\!]$. Suppose, on the contrary, that $p \nvDash [\![((\langle a, b\rangle \in f \wedge \langle a, c\rangle \in f) \supset b = c)]\!]$. Then, for some q with $p\mathcal{R}q$, $q \vDash [\![\langle a, b\rangle \in f]\!]$, $q \vDash [\![\langle a, c\rangle \in f]\!]$, and $q \vDash [\![\sim(b = c)]\!]$. Because $q \vDash [\![\langle a, b\rangle \in f]\!]$, it follows that for some g and some q' with $q\mathcal{R}q'$, $q' \vDash [\![\langle a, b\rangle \approx g\rangle]\!]$, $q' \vDash [\![\langle g \epsilon f\rangle]\!]$. Then it is easily seen that $g = \widehat{\langle n, \alpha\rangle}$ for some n and α. Because $q \vDash [\![\langle a, c\rangle \in f]\!]$ (and hence also $q' \vDash [\![\langle a, c\rangle \in f]\!]$), similar reasoning will yield $q'' \vDash [\![\langle a, c\rangle \approx \widehat{\langle m, \beta\rangle}]\!]$ and $q'' \vDash [\![\widehat{\langle m, \beta\rangle}\epsilon f]\!]$ for some q'' with $q'\mathcal{R}q''$ and some m and β. From this it easily follows that $q'' \vDash [\![\hat{n} \approx \hat{m}]\!]$. This is because $(\forall x)(\forall y)((\langle x, y\rangle = \langle a, b\rangle) \supset x = a)$ is classically true in M, hence its translate is valid in the model. Then we can use this translate twice to get both $q'' \vDash [\![a \approx \hat{n}]\!]$ and $q'' \vDash [\![a \approx \hat{m}]\!]$, from which it follows that $q'' \vDash [\![\hat{m} = \hat{n}]\!]$. Similar reasoning shows that $q'' \vDash [\![\hat{\alpha} \approx b]\!]$, and $q'' \vDash [\![\hat{\beta} \approx c]\!]$. Then it follows (by Lemma 3.2 of Chapter 17) that n and m are the same set. Thus both $q'' \vDash [\![\widehat{\langle n, \alpha\rangle}\epsilon f]\!]$ and $q'' \vDash [\![\widehat{\langle n, \beta\rangle}\epsilon f]\!]$, which can easily be seen to imply $r \vDash \widehat{\langle n, \beta\rangle}\epsilon f$ and $s \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$ for some r and s with $r\mathcal{R}s$. But, by the definition of the accessibility relation, s is an extension of r. Also, by the definition of f, $n \in \text{Dom}(r)$ and $r(n) = \alpha$. Because s is an extension of r, and $n \in \text{Dom}(s)$, it follows that $s(n) = \beta = \alpha$. Thus $\alpha = \beta$, meaning $q'' \vDash [\![\hat{\alpha} = \hat{\beta}]\!]$, meaning $q'' \vDash [\![\hat{\alpha} \approx \hat{\beta}]\!]$. This implies $q'' \vDash [\![b \approx c]\!]$, that is, $q'' \vDash [\![b = c]\!]$. But we also have $q \vDash [\![\sim(b = c)]\!]$ (by assumption), hence also $q'' \vDash [\![\sim(b = c)]\!]$, a contradiction.

(2) Now we move on to showing that formula (2) is true at p. It suffices to consider any a, b, c, and show that $p \vDash [\![((\langle a, c\rangle \in f \wedge \langle b, c\rangle \in f) \supset a = b)]\!]$. Suppose, on the contrary, that $p \nvDash [\![((\langle a, c\rangle \in f \wedge \langle b, c\rangle \in f) \supset a = b)]\!]$. Then, for some q with $p\mathcal{R}q$, $q \vDash [\![\langle a, c\rangle \in f]\!]$, $q \vDash [\![\langle b, c\rangle \in f]\!]$, and $q \vDash [\![\sim(a = b)]\!]$. Using near identical reasoning as in part (1) of this proof, it follows that for some r with $q\mathcal{R}r$ and some s with $r\mathcal{R}s$, $r \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$ and $s \vDash \widehat{\langle m, \alpha\rangle}\epsilon f$ for some n, m, and α, where $r \vDash [\![a \approx \hat{n}]\!]$ and $r \vDash [\![b \approx \hat{m}]\!]$. Because $r \vDash \widehat{\langle n, \alpha\rangle}\epsilon f$, it follows, by the definition of f, that $n \in \text{Dom}(r)$ and $r(n) = \alpha$. Similarly, $n \in \text{Dom}(s)$ and $s(m) = \alpha$. But s is an extension of r, and because both r and s are 1-1 (all forcing conditions under consideration here are 1-1 functions), it follows that $n = m$. Hence $r \vDash [\![\hat{n} = \hat{m}]\!]$, implying $r \vDash [\![\hat{n} \approx \hat{m}]\!]$. Because $r \vDash [\![a \approx \hat{n}]\!]$ and $r \vDash [\![b \approx \hat{m}]\!]$, it then follows that $r \vDash [\![a \approx b]\!]$, that is, $r \vDash [\![a = b]\!]$. But $q \vDash [\![\sim(a = b)]\!]$, meaning $r \vDash [\![\sim(a = b)]\!]$, which is a contradiction.

(3) Next, we show that formula (3) is true at p. It suffices to consider any a, and show that $p \vDash [\![(a \in \hat{\omega} \supset (\exists y)(y \in \widehat{\omega_2} \wedge \langle x, y\rangle \in f))]\!]$. Suppose, on the contrary, that $p \nvDash [\![(a \in \hat{\omega} \supset (\exists y)(y \in \widehat{\omega_2} \wedge \langle a, y\rangle \in f))]\!]$. Then it follows that, for some q with $p\mathcal{R}q$, $q \vDash [\![(a \in \hat{\omega})$ and $q \vDash [\![\sim(\exists y)(y \in \widehat{\omega_2} \wedge \langle a, y\rangle \in f)]\!]$. It follows by the usual reasoning that, for some q' with $q\mathcal{R}q'$, $q' \vDash [\![a \approx \hat{n}]\!]$ for some $n \in \omega$ such that $q' \vDash [\![\hat{n} \epsilon \hat{\omega}$. Also, $q' \vDash [\![\sim(\exists y)(y \in \widehat{\omega_2} \wedge \langle a, y\rangle \in f)]\!]$. There are two cases: either $n \in \text{Dom}(q')$, or $n \notin \text{Dom}(q')$.

Case 1) $n \in \text{Dom}(q')$. Let $q'(n) = \alpha$. In order to derive a contradiction, we would need to know that $\langle a, \hat{\alpha}\rangle \in \mathcal{D}^{\mathcal{G}}$, but we certainly do not know this. It is the case that $\langle a, \hat{\alpha}\rangle$, but this is not what allows us to prove $r \vDash [\![\langle \hat{n}, \hat{\alpha}\rangle]\!]$ for some r with $q'\mathcal{R}r$, which is what we needed.

Thus we cannot actually prove that $[\![f : \hat{\omega} \xrightarrow[\text{onto}]{\text{1-1}} \hat{\omega_2}]\!]$ is valid over $\mathcal{M}$.

Sections 3 and 4

There are multiple issues with these two sections. First, Exercise 3.1, with the given definition of f, is simply incorrect. Second, the definition of F is section 4 is incorrect. However, both of these issues can be fixed by altering the definition.

First, in order to fix Exercise 3.1, f the definition of f needs to be altered to: $p \vDash \mathrm{Op}(\hat{n}, \hat{\alpha}) \epsilon f$ iff $n \in \mathrm{Dom}(p)$ and $p(n) = \alpha$. This is shown to fix the problem in the above section. The problem of the given definition lies in the fact that, if we were to show that $[\![f : \hat{\omega} \xrightarrow[\text{onto}]{\text{1-1}} \hat{\omega_2}]\!]$ is valid over $\mathcal{M}$, we would need to show that

$$(3)\ p \vDash [\![(\forall x)(x \in \hat{\omega} \supset (\exists y)(y \in \widehat{\omega_2} \wedge \langle x, y \rangle \in f))]\!]$$

But the $\langle x, y \rangle \in f$ is the problem: we only have $p \vDash \widehat{\langle n, \alpha \rangle} \epsilon f$, and $\widehat{\langle n, \alpha \rangle}$ does not act like an ordered pair (p, nor any other world, for that matter, necesarily thinks it does). This is we why instead use $p \vDash \mathrm{Op}(\hat{n}, \hat{\alpha}) \epsilon f$. This works, as p (and any other world, for that matter), thinks that $\mathrm{Op}(\hat{n}, \hat{\alpha})$ is an ordered pair.

Second, the F in section 4 has an incorrect definition. F as currently defined isn't even a member of $\mathcal{D}^{\mathcal{G}}$, let alone $R^{\mathcal{G}}_{\omega_2+3}$. In order to fix this, change the definition of F to the following:

$$F = \{ \langle p, \mathrm{Op}(\hat{\alpha}, f_\alpha) \rangle | p \in \mathcal{G}, \alpha \in \omega_2 \}$$

In order to show why these two fixes actually work, we prove some preliminary results. The main thing we need to show is that for any $f, g \in \mathcal{D}^{\mathcal{G}}$, $[\![\langle f, g \rangle = \mathrm{Op}(f, g)]\!]$ is valid over the model. This is just an abbreviation for:

$$[\![(\forall x)(x \in \mathrm{Op}(f, g) \equiv (x = \{f\} \vee x = \{f, g\}))]\!]$$

which is in turn an abbreviation for:

$$[\![(\forall x)(x \in \mathrm{Op}(f, g) \equiv (\exists y)[x = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g))))]]\!]$$

Proving this would establish that all possible worlds think $\mathrm{Op}(x, y)$ is just $\langle x, y \rangle$.

First, here is a starter result (though not actually necessary for the proof). We prove that if $\mathrm{Op}(a, b) = \mathrm{Op}(c, d)$, then $a = c$ and $b = d$.

Proof. Suppose $\mathrm{Op}(a, b) = \mathrm{Op}(c, d)$. For simplicity's sake, let $S_1 = \mathcal{G} \times \{a\}$, $S_2 = \mathcal{G} \times \{a, b\}$, $S_3 = \mathcal{G} \times \{c\}$ and $S_4 = \mathcal{G} \times \{c, d\}$. Then $\mathrm{Up}(a, a) = S_1$, $\mathrm{Up}(a, b) = S_2$, $\mathrm{Up}(c, c) = S_3$ and $\mathrm{Up}(c, d) = S_4$. Also, $\mathrm{Op}(a, b) = \mathrm{Up}(S_1, S_2) = \mathcal{G} \times \{S_1, S_2\}$ and $\mathrm{Op}(c, d) = \mathrm{Up}(S_3, S_4) = \mathcal{G} \times \{S_3, S_4\}$.

Because $\mathcal{G} \times \{S_1, S_2\} = \mathcal{G} \times \{S_3, S_4\}$, it follows that either $S_1 = S_3$ or $S_1 = S_4$, and either $S_2 = S_4$ or $S_1 = S_3$. If the former is immediately the case, we are done. On the other hand, suppose that $S_1 = S_4$. On the one hand, suppose that $S_2 = S_3$. Then it is easily seen to follow that $f = h$. From this, and the assumption that $S_1 = S_3$, it also can easily be seen to follow that $f = k$; similarly, because $S_2 = S_3$, $g = h$ also easily follows. Then it can easily be concluded that $S_1 = S_2 = S_3 = S_4$. On the other hand, suppose that $S_2 = S_4$. If also $S_2 = S_3$, then the result follows by the above reasoning. If, on the other hand, $S_2 \neq S_3$, it immediately follows (because either $S_3 = S_1$ or $S_3 = S_2$) that $S_1 = S_3$, which also completes the proof.

□

Lemma 0.1.

$$[\![(\forall x)[x \in \mathrm{Up}(f, g) \equiv (x \approx f \vee x \approx g)]]\!]$$

is valid in $\mathcal{M}$ (this is just Exercise 4.1).

Proof. Suppose that $p \nvDash [\![(\forall x)[x \in \mathrm{Up}(f, g) \equiv (x \approx f \vee x \approx g)]]\!]$ for some p. We derive a contradiction. There must be some d such that $p \nvDash [\![[d \in \mathrm{Up}(f, g) \equiv (d \approx f \vee d \approx g)]]\!]$. There are then two cases.

Case 1) $p \nvDash [\![[d \in \mathrm{Up}(f, g) \supset (d \approx f \vee d \approx g)]]\!]$. Then it follows that for some p' with $p \mathcal{R} p'$, $p' \vDash [\![d \in \mathrm{Up}(f, g)]\!]$ and $p' \vDash [\![\sim (d \approx f \vee d \approx g)]\!]$. Because $p' \vDash [\![d \in \mathrm{Up}(f, g)]\!]$, $p'' \vDash [\![d \approx c]\!]$ and $p'' \vDash [\![c \epsilon \mathrm{Up}(f, g)]\!]$ for some c. But by the definition

of $\mathrm{Up}(f,g)$, it easily follows that $p'' \vDash [\![c \epsilon \mathrm{Up}(f,g)]\!]$ implies either $c = f$ or $c = g$. Thus either $p'' \vDash [\![d \approx f]\!]$ or $p'' \vDash [\![d \approx g]\!]$, meaning] $p'' \vDash [\![(d \approx f \vee d \approx g)]\!]$, contradicting the assumption that $p'' \vDash [\![\sim(d \approx f \vee d \approx g)]\!]$.

Case 2) $p \nvDash [\![[(d \approx f \vee d \approx g) \supset d \in \mathrm{Up}(f,g)]]\!]$. Then it follows that for some p' with $p \mathcal{R} p'$, $p' \vDash [\![(d \approx f \vee d \approx g)]\!]$ and $p' \vDash [\![\sim d \in \mathrm{Up}(f,g)]\!]$. Because $p' \vDash [\![(d \approx f \vee d \approx g)]\!]$, there is some p'' with $p' \mathcal{R} p''$ such that at least one of $p''[\![d \approx f]\!]$ or $p''[\![d \approx g]\!]$ is true at p''. Suppose, without loss of generality, that the former is true, that is, $p''[\![d \approx f]\!]$. Now, $p'' \vDash f \epsilon \mathrm{Up}(f,g)]\!]$ can easily be seen to be true. This means taht $p''[\![d \in \mathrm{Up}(f,g)]\!]$. But this contradicts the assumption that $p'' \vDash [\![\sim d \in \mathrm{Up}(f,g)]\!]$.

□

Now we prove the main preliminary result.

Theorem 0.2.

$$[\![(\forall x)(x \in \mathrm{Op}(f,g) \equiv (\exists y)[x = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))])]\!]$$

is valid over $\mathcal{M}$.

Proof. Suppose that

$$p \nvDash [\![(\forall x)(x \in \mathrm{Op}(f,g) \equiv (\exists y)[x = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))])]\!]$$

We derive a contradiction. There must be some d such that

$$p \nvDash [\![(d \in \mathrm{Op}(f,g) \equiv (\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))])]\!]$$

There are then two cases.

Case 1)

$$p \nvDash [\![(d \in \mathrm{Op}(f,g) \supset (\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))])]\!]$$

Then there is some p' with $p \mathcal{R} p'$ such that $p' \vDash [\![(d \in \mathrm{Op}(f,g))]\!]$ and $p' \vDash [\![\sim(\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))]$. Because $p' \vDash [\![(d \in \mathrm{Op}(f,g))]\!]$, it follows by the above lemma, and the definition of $\mathrm{Op}(f,g)$, that either (1) $p' \vDash [\![d \approx \mathrm{Up}(f,f)]\!]$ or (2) $p' \vDash [\![d \approx \mathrm{Up}(f,g)]\!]$. Also, by the above lemma, we know that both $p' \vDash [\![(\forall z)(z \in \mathrm{Up}(f,f) \equiv x = f)]\!]$ and $p' \vDash [\![(\forall z)(z \in \mathrm{Up}(f,g) \equiv x = f \vee x = g)]\!]$. These two things imply that, whether (1) or (2) holds, $p' \vDash [\![(\exists)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))]$ must hold, a contradiction to the assumption that $p' \vDash [\![\sim(\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))]$.

Case 2)

$$p \nvDash [\![((\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))] \supset d \in \mathrm{Op}(f,g))]\!]$$

Then there is some p' with $p \mathcal{R} p'$ such that (1) $p' \vDash [\![(\exists y)[d = y \wedge ((\forall z)(z \in y \equiv z = f) \vee (\forall z)(z \in y \equiv (z = f \vee z = g)))]$ and (2) $p' \vDash [\![\sim d \in \mathrm{Op}(f,g)]\!]$. By (1), there is some p'' with $p' \mathcal{R} p''$ and some e such that $p'' \vDash [\![[d = e \wedge ((\forall z)(z \in e \equiv z = f) \vee (\forall z)(z \in e \equiv (z = f \vee z = g)))]]\!]$. Then $p'' \vDash [\![d = e]\!]$, and there is some p''' with $p'' \mathcal{R} p'''$ such that at least one of $p''' \vDash [\![(\forall z)(z \in e \equiv z = f)$ or $p''' \vDash [\![(\forall z)(z \in e \equiv (z = f \vee z = g))]\!]$ holds. Then, by the above lemma, either From these two things it follows that either $p''' \vDash [\![e = \mathrm{Up}(f,f)]\!]$ or $p''' \vDash [\![e = \mathrm{Up}(f,g)]\!]$, meaning either $p''' \vDash [\![d = \mathrm{Up}(f,f)]\!]$ or $p''' \vDash [\![d = \mathrm{Up}(f,g)]\!]$. By the above lemma, this implies that $p''' \vDash [\![d \in \mathrm{Op}(f,g)]\!]$, contradicting the fact that it was assumed $p''' \vDash [\![\sim(d \in \mathrm{Op}(f,g))]\!]$.

□